RESEARCH REPORT / 10 NOVEMBER 2026

PHSMLP - Pooled Head MLP
A 500M-token Transformer comparison

Architecture, four-way factorial evaluation, factual-recall probes and training efficiency

Four decoder-only Transformers were trained on the same 500 million scored FineWeb-Edu next-token targets per model. The study isolates weight sharing and learned head pooling, and evaluates both language quality and contextual retrieval.

500M scored training targets per model	**13.4%** fewer weights in shared + pool
88.75 / 105.66 standard / shared + pool test perplexity	**2,000** recorded milestone health checks: OK

Finding. The shared pooled model completed training without recorded numerical collapse and learned to use its pooling path. It remains less accurate than the standard Transformer at this budget. Further controlled training and equal-budget capacity comparisons are justified; convergence or superiority has not been demonstrated.

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Abstract and research question

This experiment evaluates whether a single feed-forward network (FFN/MLP), reused across three processing passes, can retain language-model performance while a learned pooled-head pathway coordinates four independent attention heads. A 2 x 2 factorial design separates the effects of sharing and pooling. Each model processed 500M matched scored targets (2 billion total).

The standard model obtained 88.75 test perplexity, versus 105.66 for the pooled shared model. The latter uses 13.4% fewer parameters. Removing or disrupting its learned pool at evaluation worsens next-token prediction, indicating learned dependence on the additional pathway. Yet the independently trained plain shared MLP is nearly as accurate, and the standard model has stronger long-distance copying. Zero-shot fact-binding performance remains weak for all four models.

1. Model configurations

Variant	FFN weight sets	Pool	Parameters
Standard	3	No	444,992
Standard + pool	3	Yes	451,331
Shared MLP	1	No	378,816
Shared + pool	1	Yes	385,155

All configurations use three attention blocks, four heads, 64-dimensional token states, a 256-neuron GELU FFN, learned positional embeddings and tied input/output token embeddings. Attention and layer norms remain independent across depth. Only the MLP weights are tied in the shared variants.

Interpretation of the design

Pooling adds 6,339 parameters to either baseline; sharing removes 66,176. Embedding and positional weights account for 294,912 parameters in every model, reducing the percentage saving in the full network. This compares equal width, depth and training exposure, **not** equal total parameters or FLOPs.

The experiment establishes feasibility at this scale, not a general architectural ranking. The next question is whether additional training or a wider shared FFN improves the quality-efficiency frontier.

2. Architecture and mathematics

The tested architecture retains conventional causal multi-head attention and sequential pre-normalised residual updates. It does not replace attention with a gate or compress the full state into one head.

Attention and learned head pooling

For each processing pass r, let x be a token state and u = LN(x). Independent Q, K and V projections are computed together. Each head h produces a contextual value vector a_h of dimension 16 using causal scaled dot-product attention.

Baseline: b = W_A concat(a_1, a_2, a_3, a_4) + b_A.

Pool: q_p = W_q u; w_h = softmax_h(q_p · a_h / sqrt(16));
s = sum_h w_h a_h; g = sigmoid(l); A(u) = b + g (W_p s + b_p).

The normal attention output b is always retained. The learned summary s is only an additional residual contribution. Pool parameters are pass-specific; all four heads and the pooling path are differentiable and trained jointly. The initial gate logit is -1.5 (sigmoid about 0.182).

Single shared feed-forward network

z = x + A(LN_a(x))
y = z + F(LN_f(z))
F(v) = W_2 GELU(W_1 v + b_1) + b_2

The standard variants use a different F at each pass. The shared variants reuse the same W_1, W_2 and biases for all three passes; the states, attention projections and normalisation parameters still differ by pass. The output is a tied-embedding vocabulary projection after final layer normalisation.

Why this matters

The model retains full-width token representations and independent attention maps. Reusing an MLP reduces stored weights, not the number of FFN evaluations. Pooling introduces extra computation; it may change the information supplied to the MLP but does not by itself ensure reliable entity binding or depth-specific specialisation.

3. Dataset and experimental procedure

FineWeb-Edu (sample-10BT), local Parquet shard. The audit records a training-only SentencePiece tokenizer, hash-separated document splits and frozen token files; exact duplicate documents are grouped by hash. No deliberate replay of training targets is reported, although source duplicates may exist.

Setting	Recorded value
Training targets	500,000,000 per model; 2,000,000,000 total
Tokenisation / context	4,096-piece BPE / 512 tokens
Training batch / depth / heads	16 / 3 / 4
Optimiser	AdamW; weight decay 0.01; clip 1.0
Learning rate	0.00075 peak; 4M-target warmup; cosine to 0.00005
Random seed	20261109 (one seed)
Data split	500,000,001 train tokens; 2M validation and 2M test tokens
Evaluation schedule	Health/checkpoint every 1M; regular probes every 10M; expanded probes/ablations every 25M
Final language evaluation	100,000 scored positions; 14 recorded test-window aggregates
Execution	CUDA; mixed precision enabled; PyTorch 2.11.0+cu130

The four models train in interleaved 1M-target blocks on the same contiguous data windows. Validation is monitored throughout; final test text is first opened after all four reach 500M. The compact archive records all 2,000 health rows, each marked OK, and final completion metadata.

Reproducibility boundary

This report analyses the recorded audit and its supplied implementation, not an independent rerun. The source revision is recorded as a local-Parquet SHA-256; token-file and tokenizer hashes are included. The archive excludes token cache and trained checkpoints, so reproducing the final run requires the original data preparation and retraining or separately saved weights.

4. Language-model performance

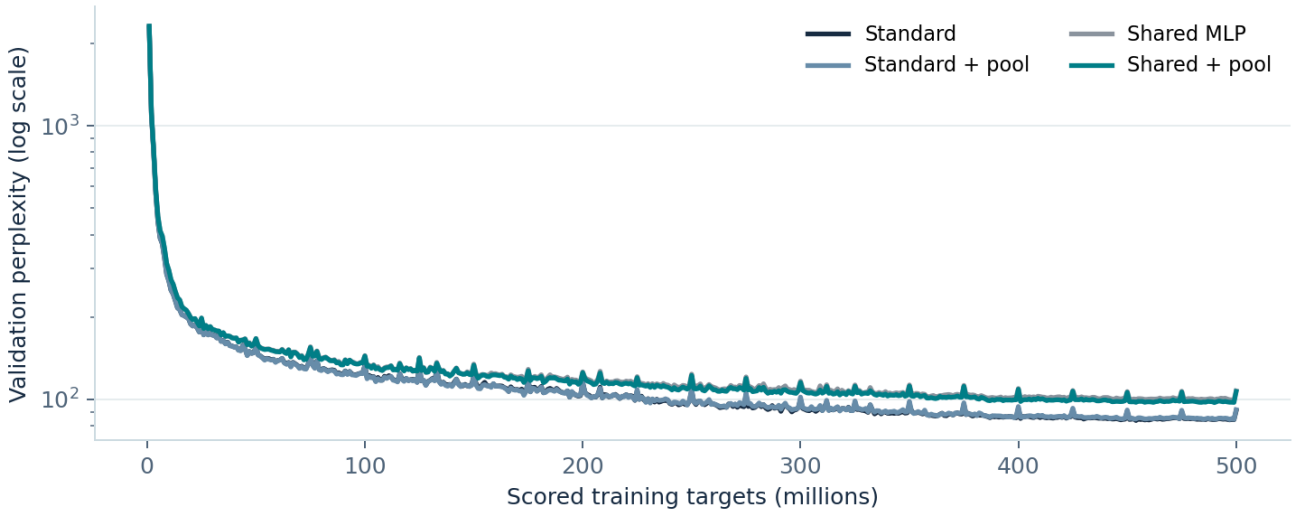


Figure 1. Validation perplexity across all 500 recorded milestones per model (log scale). Lower is better.

Final test	Standard	Std + pool	Shared	Shared + pool
NLL (nats/token)	4.4859	4.5065	4.6633	4.6602
Perplexity	88.75	90.60	105.98	105.66
Top-1 accuracy	23.95%	23.50%	21.18%	21.03%

At 500M, the shared pooled model has 19.0% higher test perplexity than standard. Its validation gap also widens: 8.3% at 100M (143.91 vs 132.92), and 17.6% at 500M (106.73 vs 90.77). Both continue to improve; the observed trajectory does not support the claim that simply extending training will close the gap.

Pooling and sharing effects

Recorded paired test-window bootstrap: pooling with shared MLP changes mean NLL by -0.0044 nats/token (95% interval [-0.0108, +0.0021]); pooling with private MLP changes it by +0.0151 [+0.0008, +0.0309]. Sharing without pooling: +0.1592 [+0.1230, +0.1900]. These are 14 window-aggregate comparisons, not estimates across independent training seeds. The full 100k-token mean differences need not equal the unweighted window means.

5. Factual recall and reasoning

Zero-shot synthetic questions ask for a single-token answer among four candidates, using 2/4/8/16/32 facts, distance gaps of 0/16/64/128/256/384 tokens, 2-4 hops, overwrites, query swaps and counterfactual edits. Nominal chance is 25%. The final main suite contains 1,408 matched questions per model; separate ablation mini-probes are excluded.

Task (n per model)	Standard	Std + pool	Shared	Shared + pool
Binding (320)	32.2%	26.6%	24.1%	25.9%
Overwrite (128)	26.6%	28.1%	29.7%	32.8%
Multi-hop (192)	27.1%	28.1%	28.1%	30.7%
Distance (384)	26.8%	28.4%	28.1%	28.1%
Counterfactual (256)	28.9%	26.2%	27.0%	27.0%
Query swap (128)	28.1%	28.9%	31.2%	24.2%
All (1,408)	28.55%	27.56%	27.41%	27.84%

The shared pooled model shows small numerical gains over plain sharing on corrections and multi-hop questions. The paired overall difference is +0.43 percentage points (95% interval -1.53 to +2.45); the combined comparison with standard is -0.71 points (-3.13 to +1.78). These intervals are consistent with no advantage. All four models score 0% on unrestricted next-token answers.

Binding by number of facts

Facts	Standard	Std + pool	Shared	Shared + pool
2	39.1%	32.8%	26.6%	31.2%
4	29.7%	28.1%	23.4%	31.2%
8	29.7%	23.4%	17.2%	18.8%
16	26.6%	21.9%	23.4%	23.4%
32	35.9%	26.6%	29.7%	25.0%

Each fact-count condition has 64 questions per model. Results are not a measure of general factual knowledge or task-trained retrieval. The probes use unusual synthetic prompts and a small vocabulary; the report does not establish dependable binding for any architecture.

6. Context copying and pooling ablations

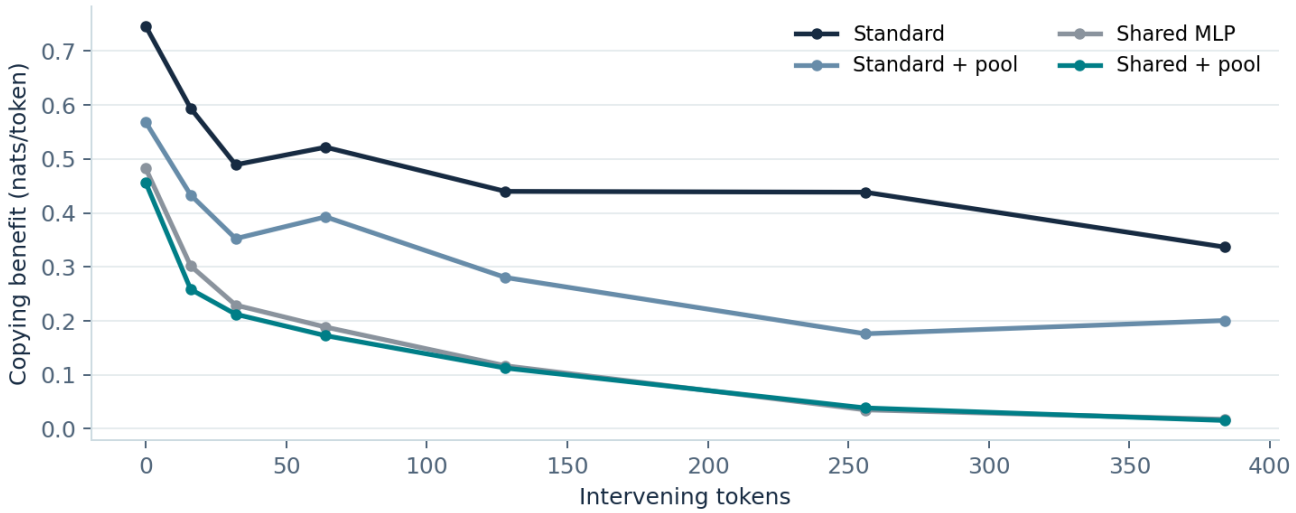


Figure 2. Mean copying benefit (control NLL minus repeated-phrase NLL), 64 examples per distance; higher is better.

The standard Transformer has the largest copying benefit at every tested gap. At 384 tokens, its benefit is 0.337 nats/token, versus 0.015 for shared + pool. This result does not support the earlier small-sample suggestion that pooling improves long-distance copying.

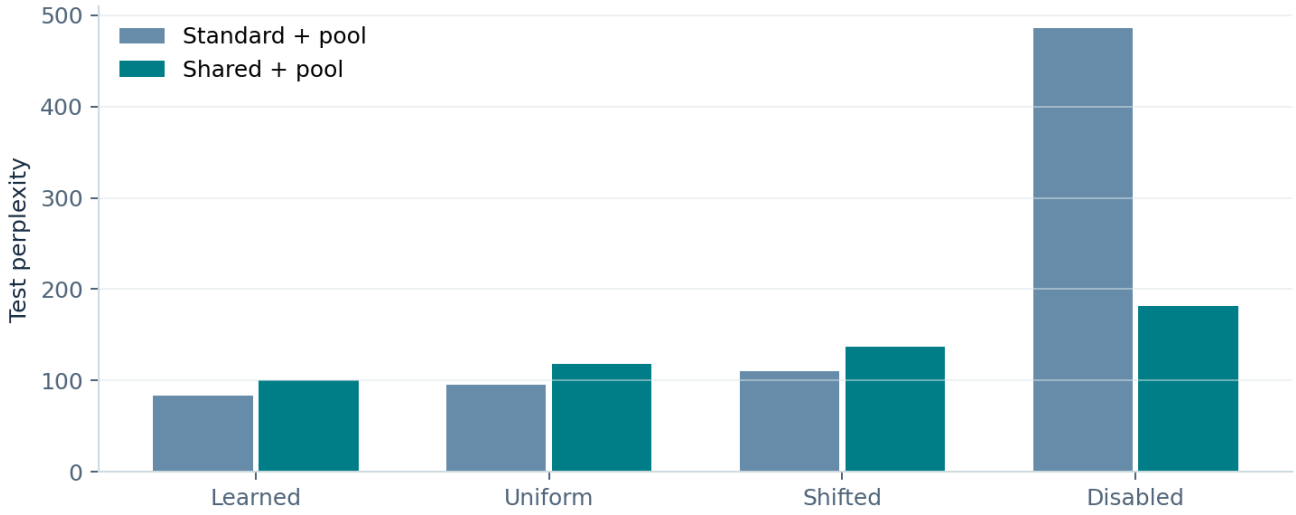


Figure 3. Evaluation-time pooling changes on an 8,192-position test subset. Lower is better.

The shared pooled model rises from 100.02 perplexity with learned pooling to 181.20 when disabled; the standard pooled model rises from 83.33 to 485.46. Learned gates reach 0.527 and 0.457 respectively. These interventions demonstrate learned dependence, not superiority to a separately trained unpooled model. Uniform and shifted weights also worsen prediction.

7. Efficiency, interpretation and next steps

Model	Median targets/s	Training minutes	Peak MB
Standard	168,919	49.4	789.9
Standard + pool	148,922	56.0	801.5
Shared	170,476	48.9	790.1
Shared + pool	150,387	55.4	801.6

Training-only timing excludes evaluation and data preparation. Shared + pool is about 11% lower in median throughput than standard and saves parameter storage, not measured GPU memory or training time. The measurements are specific to this implementation and CUDA run.

8. Conclusions and proposed follow-up

What works: all four models complete the budget; the shared architecture is trainable, the pooled path learns and contributes, and weight sharing reduces parameter count. The shared pooled model continues to improve at 500M. These are useful feasibility findings, even though the standard model leads on prediction and copying.

What remains open: more training may improve absolute quality, but there is no evidence it will close the relative gap. A matched continuation to 1B scored targets should use fresh data, the same schedule for both models and pre-specified evaluation milestones. A complementary capacity test gives the shared FFN an equal parameter budget: about 720 hidden neurons for shared + pool versus 256 in each standard FFN (estimated 445,011 total parameters); this increases FFN compute and requires a separate compute-matched comparison. Pass-specific low-rank adapters and multiple seeds would test lost specialisation.

Limitations: one seed, tiny model width, one data source, fixed context and tokenizer, differing parameter/compute budgets, small synthetic probe sets and no independent retraining. Window bootstrap intervals describe evaluation-sample uncertainty only. Neither training stability nor ablation dependence proves a general performance advantage.

Sources and audit trail

Primary: supplied RESEARCH_RESULTS_UNDER_350MB(1).zip; experiment_config.json, model_specs.json, metrics/health.csv, reports/final_results.json, paired bootstrap files, final_probe_rows.csv, copying.csv, pool_ablations.csv and research500.py. Main probe rows filtered to normal mode and first occurrence per model/probe ID, excluding extra ablation rows. Dataset: Lozhkov et al. (2024), FineWeb-Edu, doi:10.57967/hf/2497; ODC-By 1.0, subject also to Common Crawl terms. Project software: PolyForm Noncommercial 1.0.0 (see LICENSE and NOTICE.md). Format reference: github.com/rickey1990/Gated-Attention-Projection (GAP20 report). No repository publication or independent replication is claimed.